15.764 Theory of Operations Management
Final Project Proposal
An Adversarial Bandit Approach to Strategy in the National Football League

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1. Project Overview and Goal

This project evaluates National Football League (NFL) play-calling through the lens of Operations Management by framing it as a Contextual Multi-Armed Bandit problem played against an adaptive defensive adversary. We will build a mathematically optimal "Universal AI Coach" to calculate the cumulative algorithmic regret of all 32 NFL franchises. By correlating this mathematical regret with actual win totals, we aim to empirically validate our model against the hypothesis that teams diverging further from the optimal contextual policy naturally win fewer games.

2. Scope and Data Architecture

Using over 250,000 plays extracted from the public `nflverse` repository, the analysis is split into two phases to prevent look-ahead bias:

- **Training Data (2016–2020):** ∼150,000 plays to train our contextual algorithm, mathematically mapping how game state influences the Expected Points Added (EPA). We additionally leverage `nflverse` participation data to characterize empirical per-team defensive tendencies.
- **Test Data (2021–2023 Seasons):** Actual play-by-play decisions of all 32 franchises, allowing us to compare human decisions against the AI baseline across a multi-year window.

3. Theoretical Framework: Contextual Bandits & Adaptive Defense

We map play-calling to a Bandit model where Arms (A_t) are play concepts and the Reward (Y_t) is EPA. Standard algorithms fail here because they are stateless. To resolve this, we implement the **Linear Bandits (OFUL)** framework, $Y_t = X_t^\top \theta_* + \eta_t$, factoring in a Context vector (X_t: Down, Distance, Field Position) to dynamically calculate optimal policies.

Adversarial Extension: To capture the strategic nature of football, we augment the model with a Defensive Bot D_t that selects a defensive look d_t from empirical per-team distributions. The realized reward becomes $Y_t = f(X_t, A_t, d_t) + \eta_t$. The defense best-responds in expectation to the offense's tendencies, creating a simplified repeated game with simultaneous, information-limited moves.

4. Hypothesis and Statistical Validation

The Universal AI Coach defines the mathematical ceiling, making expected step-regret for a human coach strictly non-negative:

$$R_t = \mathbb{E}_{d_t \sim D_t}[\mathrm{EPA}^*_{\mathrm{AI}}(X_t, d_t)] - \mathrm{EPA}_{\mathrm{Human}}(X_t, d_t) \geq 0$$

Hypothesis: There is a strong negative correlation between a team's Total Cumulative Regret (R_T) and their regular-season wins. We expect elite teams to instinctively approximate the OFUL optimal policy against their observed defensive distributions (yielding low regret), while struggling franchises accumulate massive regret by becoming predictable and thus exploitable by adaptive defenses.

5. Project Execution Plan

- **Phases 1-2: Setup & Training:** Define state space (X_t), offensive arms (A_t), defensive arms (d_t), and reward (Y_t). Apply linear regression across 2016–2020 data to establish θ_* alongside empirical per-team defensive response distributions.
- **Phase 3: Regret Engine:** Parse the 2021–2023 test data, calculating cumulative mathematical regret for all 32 teams compared to the AI's optimal choices against each opponent's estimated defensive policy.
- **Phase 4: Validation:** Merge the regret metrics with the 2021–2023 NFL standings. Run a linear regression analysis (Regret vs. Wins) to statistically test the hypothesis.
- **Phase 5: Synthesis:** Draft the final report bridging empirical NFL data with theoretical Linear Bandit proofs to deliver a comprehensive Operations Management conclusion.